The completely depolarizing channel is not maximal in the stated less-noisy order

Disproof of Conjecture 00000009527

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Abstract

For the source's order direction, the qubit completely depolarizing channel D has a strict successor: the completely dephasing channel Z. We prove $D \preceq_{\ln} Z$ for every finite input ensemble, including noncommuting states, and $Z \not\preceq_{\ln} D$. Thus D is not even a maximal element. The obstruction survives quotienting by mutual less-noisy equivalence. Lean checks actual complex matrices, normalized Kraus families and spectral von Neumann entropy.

The order and the two actual channels

For an ensemble $\mathcal{E} = (p_i, \rho_i)$ of qubit density matrices, let

$$\chi(\mathcal{E}; \Phi) = S\left(\sum_i p_i \Phi(\rho_i)\right) - \sum_i p_i S(\Phi(\rho_i)), \qquad S(A) = -\sum_j \lambda_j(A) \log \lambda_j(A).$$

The weights are nonnegative and sum to one; $0 \log 0 = 0$. We follow the exact direction in the conjecture: $\Phi \preceq_{\ln} \Psi$ means $\chi(\mathcal{E}; \Phi) \le \chi(\mathcal{E}; \Psi)$ for every finite ensemble. Logarithms are natural, which does not change the order. The standard definitions are recalled in [2, Sections 5.2 and 5.3.3].

For any complex 2×2 matrix A, define

$$D(A) = \frac{\operatorname{tr} A}{2} I_2, \qquad Z(A) = \begin{pmatrix} A_{00} & 0 \\ 0 & A_{11} \end{pmatrix}.$$

Both are channels on the same input and output spaces. Let E_{ab} be the matrix units, $a, b \in \{0, 1\}$. A Kraus family for Z is E_{00}, E_{11}. A Kraus family for D contains two copies of $E_{ab}/2$ for every pair (a, b). For each family,

$$\sum_j K_j^* K_j = I_2, \qquad \Phi(A) = \sum_j K_j A K_j^* \quad \text{for every complex } A.$$

These identities are checked in Lean. They give trace preservation by cyclicity of trace, and complete positivity because every matrix amplification is a sum of conjugations by $I \otimes K_j$. This is the standard finite Kraus characterization, as in [2, Section 2.2].

A strict comparison, for all input ensembles

Every density state has trace one, so $D(\rho_i) = I_2/2$. Consequently $\chi(\mathcal{E}; D) = 0$ for every ensemble.

For Z, write $a_i = (\rho_i)_{00}$ and $b_i = (\rho_i)_{11}$. These entries are real and nonnegative because ρ_i is positive semidefinite. The output is $\mathrm{diag}(a_i, b_i)$, and its actual eigenvalues are a_i, b_i, with multiplicity. Put $f(x) = -x \log x$ for $x > 0$ and $f(0) = 0$. The function f is concave on $[0, \infty)$. Finite Jensen inequalities give

$$\sum_i p_i f(a_i) \leq f\left(\sum_i p_i a_i\right), \qquad \sum_i p_i f(b_i) \leq f\left(\sum_i p_i b_i\right).$$

Adding them proves $\chi(\mathcal{E}; Z) \geq 0$. No commutativity or diagonal assumption was made on the input states. Thus $D \preceq_{\ln} Z$.

Now use the equiprobable ensemble E_{00}, E_{11}. Dephasing leaves each pure state unchanged. Its individual output entropies are zero and its average output is $I_2/2$, of entropy $\log 2$. Hence

$$\chi(\mathcal{E}; Z) = \log 2 > 0 = \chi(\mathcal{E}; D),$$

so $Z \not\preceq_{\ln} D$.

Theorem 1. *The completely depolarizing channel is not a maximal element of the less-noisy relation in the direction defined in the source, even after identifying mutually equivalent channels.*

Proof. The two channel certificates and the strict comparison $D \preceq_{\ln} Z$, $Z \not\preceq_{\ln} D$ exhibit a strict successor. Therefore D is not maximal. The same strict comparison holds between their equivalence classes. □

There is also a secondary issue with a partial order on literal channels. The two reset channels $R_a(A) = \mathrm{tr}(A)E_{aa}$, $a = 0, 1$, are distinct and have Kraus families E_{a0}, E_{a1}. Both give Holevo information zero on every ensemble. Thus $R_0 \preceq_{\ln} R_1$ and $R_1 \preceq_{\ln} R_0$, despite $R_0 \neq R_1$. Quotienting repairs this secondary failure of antisymmetry, but cannot repair the strict-successor obstruction above. The separate broadcast-capacity assertion is not needed for this disproof and is not assessed here.

Formalization, checks and reproduction

Density states in Lean are actual complex 2×2 matrices with Mathlib positive semidefiniteness and trace one. Entropy is defined from Mathlib's Hermitian eigenvalues. The diagonal entropy formula is proved using spectral membership and trace invariance under unitary conjugation, so Shannon entropy is derived rather than assigned in place of von Neumann entropy.

The channel predicate uses the exact normalized finite Kraus characterization. The displayed formulas are verified on arbitrary complex input matrices. The ensemble type has arbitrary finite size, genuine density inputs and arbitrary nonnegative real weights of sum one. The Jensen argument therefore quantifies over all allowed ensembles. The explicitly constructed orthogonal ensemble witnesses strictness.

The main theorem is `depolarize_not_maximal`: it bundles the actual channel certificates and both order comparisons. A separate theorem proves failure of antisymmetry for the two actual reset maps. No general capacity formula or quantum data-processing theorem is assumed in the proof.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The whole submission is freshly built. The axiom audit contains only standard foundational axioms. There are no proof placeholders, custom axioms or external computation oracles.

The auxiliary `verify.py` uses exact rational arithmetic to reconstruct the Kraus normalization and map identities on a matrix-unit basis. Complex linearity makes these basis checks cover every complex input; Lean also proves the arbitrary-input identities directly. This script does not supply an oracle for the entropy or all-ensemble conclusions. Run `python verify.py` and `tectonic main.tex`.

The exact bilingual source, provenance, actual build logs and hashes are included. An agent developed the proof and performed a self-review; the parent separately reviewed the source, semantics, proof, build evidence and final PDF. Both internal review records are included. No external independent review or novelty is claimed.

References

[1] The Last Math Competition, Conjecture 00000009527; exact source and provenance included.

[2] J. Watrous, *The Theory of Quantum Information*, 2018, Sections 2.2, 5.2 and 5.3.3. Author-hosted draft. The cited sections supply standard definitions; the strict comparison is proved in full above.